A Science-Based Ion Transport Simulation Game
CytoTransport

K.N. Kirschner

February 1, 2026

1 Overview

This game simulates the dynamics of ion transport across a biological cell membrane (see Figure **??**). It illustrates how physical barriers and protein pore structures regulate the internal environment of a cell. The cell's environment is split into four distinct components:

Intracellular Region: Represents the protein-rich cytoplasm of the cell's interior. Rendered in gray, initially, and changes to red as the ion concentration increases.

Extracellular Region: Represents the aqueous, water-rich environment surrounding the cell. Rendered in light blue.

Cell Membrane: Represents the hydrophobic (i.e., water repelling) phospholipid bilayer that creates the cellular wall, separating the intra- and extracellular regions. Rendered in orange.

Protein Pores: Represent the tapered protein that spans the cellular wall, allowing ions to flow across. Rendered in white.

2 Game Design (i.e., the Model)

2.1 The Hydrophobic Lipid Bilayer

The phospholipid bilayer is **hydrophobic** and forms the cell wall. Due to its hydrophobicity, polar molecules (e.g., H_2O) and charged ions (e.g., Na^+, K^+, Cl^-) are unable to simply pass through it. Instead, nature created alternative pathways, including protein ion pores. Due to the water-rich intra- and extracellular environments, ions (plus polar molecules and other biopolymers) can move around.

Game Mechanic: The Lipid Bilayer Barrier

Ions interact with the lipid bilayer through collisions and are randomly scattered back.

Scientific Basis: The ion scattering represents the high energy barrier that an ion encounters when attempting to diffuse through the phospholipid's non-polar fatty acid tails. The change in the ion's trajectory reflects the scattering effect of hitting a dense, hydrophobic and heterogeneous molecular cell barrier, and the multitude of intermolecular forces that occur.

Code:

- **Bounce Logic:** `ion.vy = abs(ion.vy)` ensures a downward trajectory.
- **Randomness:** `ion.vx = random.uniform(-2, 2)` simulates slight scattering.

2.2 Protein Ion Channels

Cells need nutrients and to communicate with each other, which means that molecules and ion must pass from the extracellular to the intracellular region. While nonpolar molecules are able to diffuse through the hydrophobic lipid bilayer, many polar species must find alternative routes. As one solution, nature has created proteins that form mechanical pores that transverse the cell membrane.

Game Mechanic: Tapered Protein Wedges

Scientific Basis: The wider bottom represents the water-filled entrance of a channel. This lowers the electrostatic energy barrier, allowing ions to diffuse from the extracellular space into the channel. The channel's narrowing models the protein-specific **selectivity function**. Specific channels (e.g., Sodium Channel) are structured such that this narrow region forces ions to shed their **hydration shells** (i.e., H_2O molecules) allowing ions of certain sizes (e.g., Na^+) to migrate through.

Code:

- **Model:** The proteins the form the ion channel are rendered as **light gray** tapered wedges using `pygame.draw.polygon`.
- **Geometry:** The protein pore structures are rendered as oriented 4-point polygons to reflect the pore's geometry. The wedges are anchored at the gap boundaries (`gl, gr`), and extend into the membrane with a depth of 15px. The top points of the wedges (intracellular side) extend horizontally an additional 6px into the channel than the bottom points.

Game Mechanic: Size-Dependent Filtering

Scientific Basis: This mechanic models how pores filter ions based on their size (i.e., **Steric Exclusion**). The role that proteins channels have is diverse, ranging from nonselective to selective channels. Simplifying the complexity of ion-amino acid interactions that may occur and participation of modulators, selective ion channels can filter which ions pass as function of the pore's radius relative to the ion. If the pore constricts below a certain threshold, the ion is prohibited from moving through. In our game's model, we see that different sized ion have different difficulty getting through as the pore radius varies.

Code:

- **Filtering:** If any part of the ion's width (r) overlaps with the tapered protein wedges, a **collision** is registered.
- **Collision:** The collision logic evaluates the `ion.rect` boundaries against the pore's `current_gap`. If PoreWidth < IonSize, the probability of passage drops to zero and the ion is reflected.

Game Mechanic: Pore Opening and Closing

Scientific Basis: The protein channel's pore can increases and decrease due to structural changes in the protein's subunits (i.e., **gating**). To approximate this behavior, the proteins are translated horizontally.

Code: The protein channel's pore width is systematically changed using a sine wave:

$$\mathrm{CurrentGap} = \mathrm{BaseGap} + \sin(\mathrm{timer}) \cdot \mathrm{Amplitude}$$

2.3 Ion Characteristics

The human body requires many different ions for different reasons (e.g., Table 2). These ions have different sizes and properties. Their movement (i.e., velocity) through water is complicated, involving their size and how strongly they structure their surrounding water molecules. Ions that form well ordered and strong hydration shells essentially increase their size since they move more like a single unit. Even though cells need ions, too many (i.e., concentration - an amount per volume) can also be bad for cells and for human health.

Game Mechanic: Ion Size

Scientific Basis: In a real cell, ion channels are highly selective. A channel might be physically too narrow for a large ion to pass, or it may require the ion to shed its "hydration shell" (water molecules sticking to the charge) to fit through the selectivity filter. The simulation allows for multiple ion species, each defined by a unique **Ionic Radius**.

Code: The `Ion` class accepts a `size` parameter. Larger ions (e.g., K^+) have a wider `pygame.Rect`, making it physically more difficult to pass through the narrowing wedge of the pore during its "constricted" sine-wave phase.

Table 1: Possible ions (Scale: 7px = 1.0Å)

Ion	**Charge**	**Radius (Å)**	**Color**	**Context**
Sodium (Na^+)	$+1$	1.0	Blue	Nerve impulses
Calcium (Ca^{2+})	$+2$	1.0	Yellow	Signaling
Potassium (K^+)	$+1$	1.4	Purple	Resting potential
Chloride (Cl^-)	-1	1.8	Green	Inhibition

Game Mechanic: Ion Velocity Scaling

Scientific Basis: The game assigns a unique velocity vector to each ion species. A particle's movement can be explained by the closely related **Jones-Dole Relation** and the **Stokes-Einstein Relation**. The Jones-Dole relationship expresses how an ion changes the local water structure (i.e., how tightly bound its hydration shell is), while the Stokes-Einstein relationship measure how the ion moves through water.

The B-coefficient determines the ion's impact on the local solvent structure [2] as expressed in the **Jones-Dole Relation**:

$$\eta_{rel} = 1 + A\sqrt{c} + Bc$$

where η_{rel} is the relative velocity, A and B are coefficients, and c is the molar concentration. The large B-coefficient values indicate smaller velocity.

The **Stokes-Einstein Relation** is defined as:

$$D = \frac{k_B T}{6\pi\eta r}$$

where D is the diffusion coefficient, k_B is the Boltzmann constant, T is the temperature, η is the dynamic viscosity and r is the hydrodynamic (Stokes) radius of the particle.

Ions with a **large positive B-coefficient** (like Na^+ and Ca^{2+}) strongly coordinate water molecules, increasing the local viscosity and effectively increasing their **Stokes Radius**. This is why these ions move more slowly in the simulation compared to "structure breakers" like K^+, which have negative B-coefficients and lower the local resistance to flow.

Code:

Table 2: Ions and their aqueous B-coefficients at 25 ^oC ($dm^3 \cdot mol^{-1}$) [2] and Radius (Å) [1].

Ion	B-Coefficients	Radius	Context
Potassium (K^+)	-0.009	1.41	less ordered local structure; fast velocity
Chloride (Cl^-)	-0.005	1.80	less ordered local structure; fast velocity
Sodium (Na^+)	0.085	0.97	high ordered local structure; medium velocity
Calcium (Ca^{2+})	0.284	1.03	high ordered local structure; slow velocity
Magnesium (Mg^{2+})	0.385	0.70	high ordered local structure; slowest velocity

Game Mechanic: Intracellular Ion Concentration

Scientific Basis: The intracellular environment is filled with organelles, proteins, and cytoskeletal structures, which is represented by the initial gray neutral tone, distinct from the light blue "water" in the extracellular region. As the intracellular ion concentration increases it can have bad effects for the cell and human health. For example, increasing the ion concentration can also cause the cell's **Osmotic Pressure** to increase, resulting in pulling in more water into the cell (a hypotonic situation). Increased ion concentration can also be described by their **Chemical Potential** (μ). In high-concentration systems, the number of molecular collisions per second increases, effectively increasing the "force" pushing particles back toward the pores.

Code:
Intracellular (Dynamic): To provide immediate visual feedback about the magnitude of the intracellular ion concentration, the intracellular background color changes. The top half starts as a **gray**, indicating a neutral internal environment. As the ion concentration rises, background transitions towards red.

Extracellular (Static): The bottom half of the screen remains light blue, reflecting a stable, water-rich environment.

Osmotic Pressure: The simulation tracks a `pressure_multiplier` (Π) which acts as a scalar for the velocity vectors of internal particles. While the background color provides visual feedback of the concentration gradient, the physical manifestation of this gradient is modeled by a Pressure Factor calculated using:

$$\Pi = 1.0 + \left(\frac{n_{intracellular}}{25.0}\right) \tag{1}$$

When an ion $\vec{v}$ collides with a boundary, the new velocity $\vec{v}'$ is determined by:

$$\vec{v}' = -(\vec{v} \cdot \Pi) \tag{2}$$

Table 3: Scaling and Pressure Multiplier as a Function of Ion Concentration.

Ion Count	**Multiplier (Π)**	**Concentration State**
0	1.00×	Zero (Light Gray)
10	1.40×	Low (Pale Pink)
25	2.00×	Medium (Salmon Red)
50	3.00×	High (Deep Red)

3 Future Development & Feature Ideas

- **Ion Selectivity:** Introduce different radii for ions (e.g., Ca^{2+} vs Na^{+}).
- **Relative Position:** Move the cell wall along the y-axis.
- **Modulators:** Add molecules that changes the pore dynamics (e.g., closed, open). Maybe change the function that regulates the opening and closing too.

- **Extracellular Proteins:** Add extracellular proteins that float and block the pores access briefly.
- **Intracellular Proteins:** add intracellular organelles that consume the ions.
- **Multiple Ion Types:** Add the ability to cycle through different ions (and thus, their sizes).
- **Ion nature dependency:** Provide a specific ”Score Multiplier” logic in the Python code so that successfully transporting a slow, bulky chloride ion gives the player more points than a fast sodium ion? Maybe also think about the role of the formal charge, or the effects of hydration (e.g., sodium ion has a tight solvation shell, making its effect size and speed slower).
- **Electrochemical Gradients:** Program pores with a static charge to attract or repel specific ion types.
- **Active Transport:** Add a “Pump” mechanic requiring energy (ATP) to move ions against the pressure gradient.

References

[1] Marcus, Yizhak. ”Ionic radii in aqueous solutions.” Chemical Reviews 88, no. 8 (1988): 1475-1498.

[2] Jenkins, H. Donald B., and Yizhak Marcus. ”Viscosity B-coefficients of ions in solution.” Chemical reviews 95, no. 8 (1995): 2695-2724.